Image DNA

Intelligent Image Authenticity & Forgery Detection System

Project Documentation

Part 1: Project Planning & Management

Part 2: Literature Review

Part 3: Requirements Gathering

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Part 1: Project Planning & Management

1.1 Project Overview

Digital images are now trivial to forge. Graduation certificates, official documents, ID card photos and personal portraits can be edited in Photoshop (changing a number or a name) or regenerated with AI (face swapping, inpainting, fully synthetic faces). The human eye can no longer reliably tell an authentic image from a forged one, and there is no everyday tool that lets an ordinary person or an organisation check an image quickly and credibly.

Image DNA is a layered verification platform that works like an inspection checkpoint for images. Every image submitted passes through **two complementary verification barriers**, an invisible watermark check and an AI and deepfake detector, and the evidence from both is combined into one explainable verdict.

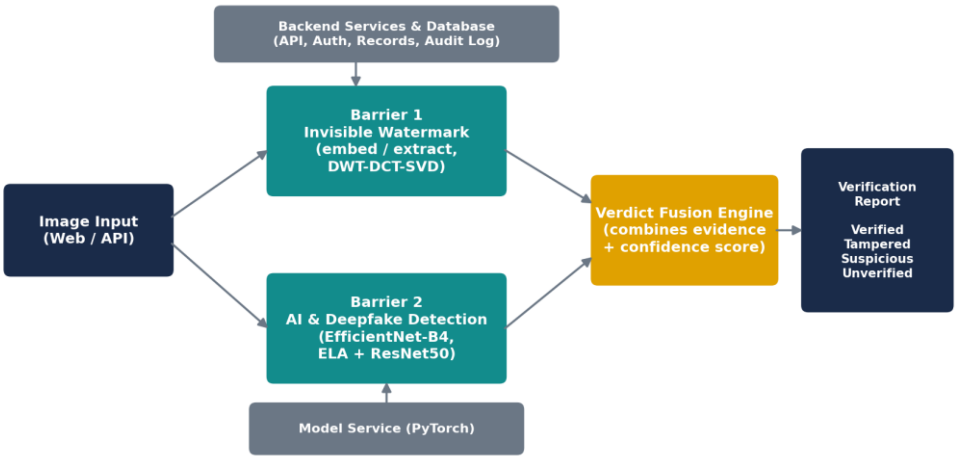


Figure 1: High-level architecture of Image DNA (two barriers + verdict fusion).

1.2 Problem Statement

- Image editing and generative-AI tools are widely available, cheap and increasingly realistic.
- Visual inspection and simple metadata (EXIF) checks are unreliable; metadata is trivially stripped or edited.
- Single-technique tools fail: a watermark can be removed or damaged, and an AI detector can be fooled or can misjudge new generators. Each technique needs a second signal to back it up.
- Organisations (universities, HR, banks, notaries) lack an accessible way to verify the authenticity of images submitted to them.

1.3 Vision, Goals and Objectives

Vision: make image authenticity checkable by anyone in seconds, with explainable evidence, measured performance, and honesty about its confidence at all times.

ID	Objective (SMART)	Measure of completion
O1	Implement Barrier 1: invisible watermark embedding and extraction that survives common benign operations (JPEG recompression, resize, mild noise).	Watermark bit-accuracy >= 95% after JPEG Q70 and 50% resize on the test set; PSNR >= 38 dB (invisible).
O2	Train and evaluate Barrier 2: a deep-learning model that detects manipulated images, face swaps and AI-generated images without having seen the original.	AUC >= 0.95 on in-distribution test data; documented performance on unseen generators / datasets.
O3	Build a fusion engine that outputs a clear verdict (Verified / Tampered / Suspicious / Unverified) with a confidence score and explanation.	Verdict returned in < 10 s per image on the reference server; report explains each barrier's contribution.

ID	Objective (SMART)	Measure of completion
O4	Deliver a web client and REST API (authentication, history, PDF report, Arabic and English) that run both barriers and show the report.	First-time users complete a verification without training (>= 90% task success); API documented with OpenAPI; 50 concurrent verifications in the load test.

1.4 Project Scope

In scope

- Still images (JPEG/PNG) of faces, ID cards, certificates and general documents.
- Web client, REST backend, database of issued watermark records, and a model-serving service.
- Detection of: watermark loss or damage, splicing / copy-move / retouching, face swap, and AI-generated faces and images.
- Verification report (on screen and as PDF) with per-barrier evidence.

Out of scope (version 1)

- Video and audio deepfake detection (future work).
- Cryptographic signing and provenance (hash registry, digital signatures, C2PA-style manifests) and mobile secure capture with device attestation (future work).
- Legal certification or admissibility of results in court.
- Detection guarantees against adaptive adversaries who know the model internals.
- Verification of the real-world truth of what a photo depicts (we verify the image, not the event).

1.5 Solution Overview: Two Barriers and a Fusion Engine

Barrier	Technique	What it answers	Key limitation (and how we cover it)
1. Invisible Watermark	Frequency-domain (DWT-DCT-SVD) embedding and extraction; optional neural encoder/decoder as a stretch goal	Was this image issued by our system, and is the mark intact?	Can be damaged by heavy edits, screenshots or cropping. Covered by Barrier 2 and by reporting "watermark damaged" instead of accusing.
2. AI & Deepfake Detection	EfficientNet-B4 (transfer learning) with MTCNN/RetinaFace face extraction; ELA + ResNet50 for pixel-level tampering	Do internal image traces show manipulation or synthetic generation, even for never-seen images?	Probabilistic, may generalise poorly to new generators. Covered by confidence reporting, cross-dataset evaluation and Barrier 1.

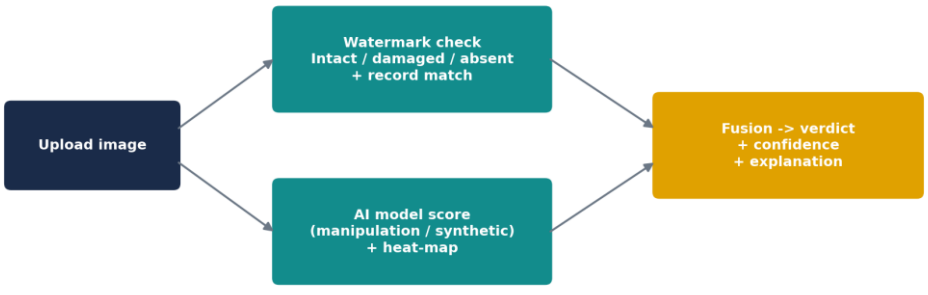


Figure 2: Verification flow and verdict fusion.

> **Design principle (honesty by design):** an image that carries an intact system watermark matching an issued record, and shows no sign of manipulation, is reported as Verified. An image without that evidence can never be declared "100% authentic"; the system reports the strength of the evidence instead (Unverified or Suspicious with a confidence score). This avoids false assurance, which is the worst failure mode for a verification product.

1.6 Selected Models and Techniques

The technical core of the project is built on the techniques below. Barrier 1 uses classical signal processing; Barrier 2 uses deep learning and image forensics.

- **Signal-based watermarking (Barrier 1):** frequency-domain transformations (DWT-DCT-SVD) for robust, invisible payload embedding and extraction.
- **Face manipulation and deepfake detection (Barrier 2):** EfficientNet-B4 [18] as the core backbone with transfer learning, preceded by MTCNN [20] / RetinaFace [21] for automated face extraction and alignment. Trained and evaluated on benchmark datasets (Celeb-DF / FaceForensics++).
- **Pixel-level tampering detection (Barrier 2):** Error Level Analysis (ELA) for compression-artifact inspection, combined with a ResNet50 [19] classifier to detect local splicing, copy-move and Photoshop edits, using the CASIA dataset (Columbia for cross-dataset testing).
- **Explainability:** Grad-CAM heat-maps for the face model and ELA maps for the tampering model, shown in the report.

1.7 Development Methodology

The team follows an **Agile / Scrum-style** process with one-week sprints (4 sprints over 4 weeks). Each sprint ends with a working increment and a short demo to the supervisor. Work is tracked with GitHub Issues and a GitHub Projects board (To do / In progress / Review / Done). Rules: feature branches, pull-request review by at least one other member, a short daily sync, and CI running unit tests on every merge to main.

1.8 Work Plan and Timeline (4 Weeks)

Task	W1	W2	W3	W4
Requirements, API contract, design freeze				
Datasets and preprocessing				
Barrier 1: watermark engine and robustness tests				
Barrier 2a: face deepfake model (EfficientNet-B4)				
Barrier 2b: tampering model (ELA + ResNet50)				
Backend, database and REST API				
Web client and PDF report				
Fusion engine and verdict rules				
Integration and Docker deployment				
Testing, evaluation and security review				
Final documentation, demo, delivery				

Figure 3: Project Gantt chart (weeks W1 to W4).

Phase	Weeks	Main deliverables
Design and data preparation	W1	Requirements and API contract frozen; database schema; datasets prepared (Celeb-DF / FaceForensics++ face crops, CASIA / Columbia ELA images); watermark prototype; baseline models; project skeletons (backend, web client, CI).
Core implementation	W2	Watermark engine meeting the robustness targets; face-deepfake and tampering models v1 trained and evaluated; REST endpoints, authentication and database; web upload and report screens.

Phase	Weeks	Main deliverables
Integration	W3	Fusion engine; model service with Grad-CAM / ELA heat-maps; full verification pipeline; web client connected to the API; Docker Compose deployment.
Evaluation and delivery	W4	Evaluation report (AUC, F1, FPR, cross-dataset); load and security tests; usability test; final documentation; demo video; release v1.0.

Milestones

Milestone	Week	Exit criterion
M1: Requirements, design and baselines	End of W1	Supervisor sign-off on documentation Parts 1-3; API contract frozen; baseline watermark prototype and baseline models run.
M2: Barrier 1 working, AI models v1	End of W2	Barrier 1 meets O1 on the test set; both Barrier 2 models trained with first in-distribution metrics.
M3: Integrated beta	End of W3	Web client calls the backend and shows the fused verdict end to end.
M4: Final release	End of W4	All acceptance tests pass; Barrier 2 meets O2 (or the gap is documented); documentation delivered.

1.9 Team Structure and Responsibilities

Every member owns one technical component end to end (design, implementation, tests and documentation) and no member has planning-only duties. Sprint planning and the weekly demo are shared and rotate between members.

Member	Component owned	Main technical tasks
Abdullah Mahmoud Fathy Fadl	Barrier 2b: tampering detection; fusion engine	Prepare CASIA v2 / Columbia; generate ELA images; train and calibrate the ResNet50 model; ELA heat-maps; implement fusion rules, confidence score and explanation text; fusion unit tests.
Ahmed Ali Farik Elkazaz	Backend, database, deployment	FastAPI REST API and OpenAPI docs; PostgreSQL schema; authentication, RBAC and API keys; file validation; audit log; model-service integration; Docker Compose and GitHub Actions CI; load and security tests.
Mazen Soudy Abd Elhalim Morsy	Barrier 1: watermark; PDF report	DWT-DCT-SVD embed and extract; payload encoding; robustness suite (JPEG, resize, noise); PSNR / SSIM measurement; watermark record matching; PDF report generator.
Shahd Mohammed Mostafa Elhewehi	Barrier 2a: face deepfake detection	Frame extraction from Celeb-DF / FaceForensics++; MTCNN / RetinaFace face crops; train EfficientNet-B4; Grad-CAM; cross-dataset evaluation; model export and inference wrapper.
Shereen Mohamed Abdelstar Mostafa	Web client and end-to-end QA	Web client (upload, issue, report, history); Arabic / English with RTL layout; accessibility checks; end-to-end test automation; usability test; demo video.

Review and ownership: each component owner is responsible for the design, implementation, tests and documentation of their component. Backup reviewers: Shahd and Abdullah review each other's models; Ahmed and Shereen review the API and web integration; Abdullah reviews the watermark module as the consumer of its output. The supervisor is consulted at every milestone.

1.10 Risk Management

#	Risk	Prob.	Impact	Mitigation / contingency
R1	AI model generalises poorly to unseen generators or datasets	High	High	Train on multiple datasets/generators; evaluate cross-dataset; report confidence, not binary truth; schedule retraining; fall back on Barrier 1.
R2	Insufficient or low-quality training data, class imbalance	Med	High	Use public benchmark datasets; augmentation; generate additional fakes with open tools; balanced sampling.
R3	Limited GPU/compute for training	Med	Med	Use transfer learning with pre-trained backbones (EfficientNet-B4, ResNet50); free cloud GPU tiers; reduce image size / batch size or use mixed precision if memory is limited.
R4	Watermark destroyed by screenshots, heavy compression or cropping	High	Med	Define supported operations explicitly; combine with Barrier 2 evidence; clear status "watermark damaged" instead of a false alarm.
R5	False positives wrongly accuse a genuine image	Med	High	Tune thresholds for a low false-positive rate; three-level verdict with a "Suspicious" band; human review guidance.
R6	Privacy / legal concerns when handling IDs and faces	Med	High	Data minimisation; encryption at rest and in transit; retention limits; explicit consent; no training on user uploads without consent.
R7	Team member unavailable / unequal workload	Med	Med	Backup reviewer for each component; daily sync; documented code; flexible task reallocation.
R8	Scope creep (video, extra document types)	Med	Med	Scope frozen at M1 (end of W1); extras tracked as future work.
R9	Integration problems between modules	Med	Med	API contract agreed in W1; integration tests from W2; Docker Compose environment.
R10	Adversarial attacks on the detector	Low	Med	Document as limitation; test basic attacks (resize, blur, JPEG); plan adversarial training as future work.
R11	The 4-week schedule is too tight	High	High	Must-have requirements first, Should/Could as stretch; parallel ownership of components; pre-trained backbones and dataset subsets; weekly demo to detect delays early.
R12	A valid watermark is copied onto another image (no cryptographic binding)	Low	Med	Match payload against a database record; Barrier 2 still analyses the image; documented limitation; stronger binding listed as future work.

1.11 Key Performance Indicators (Success Metrics)

Area	KPI	Target
Detection quality	AUC / F1 of Barrier 2 on held-out test set	AUC >= 0.95; F1 >= 0.90
Generalisation	AUC on unseen dataset/generator	Documented; target >= 0.80
False positives	Genuine images flagged as fake	<= 5% at the chosen operating point
Watermark robustness	Bit accuracy after JPEG Q70 / 50% resize	>= 95%
Imperceptibility	PSNR / SSIM of watermarked vs original	>= 38 dB / >= 0.97
Performance	End-to-end verification time	< 10 s (95th percentile)
Usability	Task success in user test (n >= 10), SUS	>= 90% success; SUS >= 75
Project delivery	Milestones met on time; test coverage of core modules	>= 80% on time; >= 70% coverage

1.12 Assumptions, Constraints and Proposed Technology Stack

- **Assumptions:** access to public datasets; availability of a GPU (local or cloud); the supervisor approves the scope.
- **Constraints:** 4-week academic schedule, student budget (open-source tools only), limited training compute.
- **Proposed stack (to be confirmed at design review):** Python, PyTorch, OpenCV; FastAPI backend; PostgreSQL; Redis (optional); Docker; React web client; GitHub Actions for CI.

Part 2: Literature Review

2.1 Review Method

We searched for existing standards, research and products that address image authenticity, grouped into four families: (1) provenance and cryptographic signing, which we review as context only because it is outside the scope of this project, (2) watermarking, (3) classical image forensics, and (4) learning-based manipulation and deepfake detection. Families 2 to 4 correspond to the techniques used in Image DNA. For each family we describe how it works, what it does well, and where it falls short, then compare it with Image DNA. Sources are listed in section 2.10.

2.2 Provenance Approaches (Context Only)

C2PA / Content Credentials. The Coalition for Content Provenance and Authenticity publishes an open specification for attaching a cryptographically signed manifest to media. The manifest records origin, edits, tools used and whether generative AI was involved, and each conforming edit adds a new manifest referencing the previous one, forming a provenance chain [1][2]. Strengths: open standard, industry adoption, tamper-evident. Weaknesses: it proves provenance, it does not detect fakes; most images carry no credentials, and many pipelines and platforms strip metadata, so a manifest often does not survive to the viewer; the C2PA ecosystem itself defines soft bindings to recover credentials for this reason [3]. Absence of credentials is not evidence of forgery.

Secure-capture platforms (e.g., Truepic). Commercial products sign images at the moment of capture inside a controlled camera pipeline, so the image is trusted from creation [23]. Strengths: strong trust at source. Weaknesses: only helps for images captured with the vendor's tooling; useless for the large existing pool of unsigned images.

Image DNA does not implement provenance signing or secure capture in this version. These approaches are complementary to our work and are listed as future work.

2.3 Watermarking

Classical frequency-domain watermarking (DWT, DCT, SVD) hides bits in transform coefficients and is fast, simple and fairly robust to compression [4]. Neural watermarking trains an encoder/decoder pair end to end with simulated distortions: HiDDeN [5] and StegaStamp [6] (robust even to printing and photographing) are representative. Generative-model watermarks such as Stable Signature [7] and Google's SynthID [22] mark AI-generated images at creation. Strengths: invisible, no external lookup needed to detect the mark. Weaknesses: only identifies images that were marked; can be attacked by regeneration, heavy cropping or screenshots; marking is cooperative, not universal.

2.4 Classical Image Forensics

Error Level Analysis (ELA) and tools such as FotoForensics highlight regions whose JPEG compression history differs from the rest of the image [8]; related techniques include noise-residual (PRNU) analysis, double-JPEG detection, copy-move detection and lighting inconsistency analysis. Strengths: no training needed, interpretable heat-maps. Weaknesses: need expert interpretation, easy to defeat by re-saving or resizing, high false-positive rates on images that were legitimately recompressed, and weak against modern AI-generated images.

2.5 Learning-Based Manipulation and Deepfake Detection

FaceForensics++ [9] introduced a large benchmark of face manipulations (Deepfakes, Face2Face, FaceSwap, NeuralTextures) and showed that CNN detectors such as XceptionNet [10] perform strongly on seen manipulation types and compressed data. Celeb-DF [14] added higher-quality fakes and the DFDC challenge dataset [11] added scale and diversity. A recurring finding is poor generalisation: detectors trained on one generator degrade on others. Work such as UnivFD [12] (features from large pre-trained models) and DIRE [13] (reconstruction error from diffusion models) tackle generalisation to GANs and diffusion models.

Commercial detectors (e.g., Sensity [24], Hive [25], Intel FakeCatcher for video [26]) offer APIs or enterprise platforms, but they are closed, produce a probability only, and publish limited independent evidence of robustness. Strengths: can analyse a never-seen image. Weaknesses: probabilistic, dataset bias, an arms race with generators, vulnerable to adversarial perturbation and to compression/resizing.

2.6 Comparison Table

Solution / family	Approach	Strengths	Weaknesses	Relation to Image DNA
C2PA / Content Credentials [1][2]	Signed provenance manifest	Open standard, tamper-evident, edit history	No fake detection; metadata often stripped; needs tool adoption	Complementary; outside project scope (future work)
Truepic-style secure capture [23]	Sign at capture	High trust at source	Vendor-specific; only new, captured images	Not implemented (future work)
Classical watermarking [4]	DWT/DCT/SVD hidden bits	Simple, invisible, fast	Fragile to heavy edits and screenshots	Barrier 1 baseline
Neural watermarking [5][6][7]	Learned encoder/decoder	More robust, covers AI outputs	Cooperative; needs training; attackable	Optional Barrier 1 upgrade
ELA / FotoForensics [8]	Compression-artifact analysis	No training, visual	Expert needed; many false positives	Input to Barrier 2b and heat-map visualisation
FaceForensics++ detectors [9][10]	CNN on face manipulations	Strong on seen methods	Weak cross-generator generalisation	Core of Barrier 2a with improved evaluation
UnivFD / DIRE [12][13]	Pre-trained features / diffusion reconstruction	Better generalisation	Heavier compute; newer, less mature	Candidate backbones for Barrier 2
Commercial detectors [24][25][26]	Closed AI APIs	Polished, scalable	Black box; cost; probability only	Benchmark / comparison only
Image DNA	2 barriers fused into one verdict	Complementary evidence; explainable	Student-scale data/compute; AI layer remains probabilistic	-

2.7 Gap Analysis and What Image DNA Adds

- Existing tools are single-technique; each fails exactly where another succeeds. Image DNA fuses two complementary signals (watermark and AI/forensic evidence) so that one weak signal does not decide the outcome.
- Watermark-only tools say nothing about images that were never marked, and detectors say nothing definitive about images the issuer did release. Image DNA handles both: issuer-mark evidence when the image was issued by our system, and statistical evidence when it was not.
- Most solutions return a bare score or a binary label. Image DNA returns an explainable report (per-barrier evidence, heat-maps, confidence).
- The focus on identity documents, certificates and faces in an accessible web experience for organisations such as universities and HR is under-served by research prototypes.

2.8 Lessons Applied to Our Design

- Evaluate cross-dataset (train on one source, test on another), because in-distribution scores overstate real-world performance [9][12].
- Use pre-trained backbones (EfficientNet/Xception/ViT) with transfer learning to cope with limited data and compute.

- Never treat a missing watermark or missing provenance as proof of forgery [3]; map evidence to a graded verdict.
- Simulate real-world distortions (JPEG, resize, noise, screenshot) when training and testing both the watermark and the detector.
- Publish limitations openly: this builds trust and prevents misuse of the verdict.

2.9 Candidate Datasets

Dataset	Content	Use in project
FaceForensics++ [9]	Real and manipulated face videos (frames extracted)	Face-swap / reenactment training and testing
Celeb-DF [14]	Celebrity real and deepfake videos (higher visual quality)	Face-manipulation training and cross-dataset evaluation
DFDC [11]	Large deepfake video dataset	Additional face-manipulation data, diversity
CASIA v2 [15] / Columbia [16]	Splicing and copy-move image forgeries	ELA + ResNet50 pixel-level tampering detection
FFHQ [17] + generated sets (StyleGAN, diffusion)	Real faces and AI-generated counterparts	Synthetic image detection, cross-generator tests
Team-made document set	Own certificates / ID mock-ups, edited in Photoshop	Domain-specific testing; no real personal data

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Part 3: Requirements Gathering

3.1 Stakeholders and Their Needs

Stakeholder	Description	Main needs
Individual users	People checking a photo, certificate or ID received from someone else, or protecting their own images	Fast, simple check; clear verdict in plain language; privacy of uploads
Document issuers	Universities, training centres, companies issuing certificates	Watermark issued documents; record of what was issued; audit trail
Verifiers	HR departments, banks, notaries, recruiters, government offices	Reliable verification; explainable report to attach to a case file; API integration; bulk checks
Journalists / fact-checkers	Verify images before publication	Evidence per technique, heat-maps, confidence level
System administrator	Operates the platform	Monitoring, user management, backups, model updates
Project team / supervisor	Builds and evaluates the system	Maintainable code, measurable results, documentation
Attackers (negative stakeholder)	Try to bypass or abuse the system	Considered in security requirements and threat model

3.2 Use Cases

ID	Use case	Actor	Summary
UC1	Issue (watermark) an image/document	Issuer	Issuer uploads an original; the system embeds an invisible watermark, stores the record and returns the protected image
UC2	Verify an image	User / Verifier	User submits an image; the system runs both barriers and returns a verdict report
UC3	View and export report	User / Verifier	User opens per-barrier evidence and exports a PDF report
UC4	Manage users and models	Admin	Role management, deploy a new model version
UC5	Verify via API	Organisation system	Third-party system calls the REST API with an API key and receives a JSON verdict

3.3 Functional Requirements

Priority uses MoSCoW: M = Must, S = Should, C = Could.

ID	Requirement	Priority
FR-01	The system shall accept JPEG and PNG images up to a configurable size (default 15 MB) from the web client and the API.	M
FR-02	The system shall validate file type and integrity and reject unsupported or corrupt files with a clear message.	M
Barrier 1: Invisible watermark		
FR-10	The system shall embed an invisible payload (issuer ID + record ID) into an image when it is issued.	M
FR-11	The system shall extract the watermark from a submitted image and report intact / partially damaged / absent.	M
FR-12	The watermark shall remain readable after JPEG recompression (Q >= 70) and moderate resizing.	M

ID	Requirement	Priority
FR-13	The system shall store each issued record (payload, issuer, timestamp) and match an extracted payload against it.	M
FR-14	The system should localise regions where the watermark is missing (indicating local editing).	C
Barrier 2: AI & deepfake detection		
FR-20	The system shall classify an image as authentic vs manipulated/synthetic and return a probability score without needing the original.	M
FR-21	The system shall detect face swaps / face manipulation in images containing faces.	M
FR-22	The system shall detect pixel-level tampering (splicing, copy-move, retouching) using ELA and a ResNet50 classifier.	M
FR-23	The system should detect fully AI-generated images (GAN/diffusion) as a separate score.	S
FR-24	The system shall generate a heat-map highlighting regions that influenced the decision, and expose model version and evaluation metadata with each result.	S
Fusion and reporting		
FR-30	The fusion engine shall combine barrier outputs into one verdict: Verified, Tampered, Suspicious, or Unverified, with a confidence score.	M
FR-31	The report shall list each barrier's result and explain in plain language why the verdict was reached.	M
FR-32	The user shall be able to export the report as PDF and view verification history.	S
Web client and platform		
FR-40	The web client shall let users upload an image, verify it, view the report; and let issuers issue (watermark) images.	M
FR-41	The system shall provide user authentication, role-based access (user, issuer, admin) and API keys.	M
FR-42	All communication shall use TLS 1.2+ and sensitive stored data shall be encrypted.	M
FR-43	The admin shall be able to manage users and deploy new model versions.	S
FR-44	The system shall log every verification and administrative action in a tamper-evident audit log.	S
FR-45	The system shall support Arabic and English user interfaces.	S

3.4 Verdict Definitions

Verdict	Condition (summary)	Meaning for the user
Verified	Watermark intact and its payload matches an issued record, and Barrier 2 scores are below the low threshold	The image carries a valid mark from the issuer and shows no sign of manipulation (decision support, not a legal proof)
Tampered	Watermark found but locally damaged together with a high Barrier 2 score, or Barrier 2 score above the high threshold	Strong evidence that the image was modified or generated
Suspicious	Barrier 2 or forensic score between the low and high thresholds, or watermark damaged without other evidence, or an extracted payload with no matching record	Treat with caution; seek the original or manual review
Unverified	No watermark found and Barrier 2 scores below the low threshold	Nothing proves it authentic and nothing indicates forgery

The low and high thresholds are chosen on the validation set to keep the false-positive rate at or below 5%.

3.5 Non-Functional (Quality) Requirements

ID	Category	Requirement	Measure
NFR-01	Performance	Verification of one image shall complete quickly under normal load	< 10 s at 95th percentile (images up to 5 MB); watermark extraction < 3 s
NFR-02	Accuracy	Detection quality shall meet the targets in Part 1	AUC >= 0.95 (in-distribution); FPR <= 5%; results reported with confidence intervals
NFR-03	Security	Data encrypted in transit and at rest; passwords and API keys stored hashed; protection against OWASP Top 10	TLS 1.2+, AES-256 at rest, no critical findings in security review
NFR-04	Privacy	Data minimisation; uploaded images deleted after a configurable retention period unless the user chooses to store them; no use for training without consent	Retention policy enforced and tested
NFR-05	Usability	A first-time user shall verify an image without training; verdicts in plain language with colour and icon, not colour only	>= 90% task success; SUS >= 75
NFR-06	Reliability	Graceful degradation: if one barrier fails, the other still runs and the report states which one failed	Availability >= 99% during demo period; no crash on invalid input
NFR-07	Scalability	Stateless API and separate model service allow horizontal scaling	Supports 50 concurrent verifications in load test
NFR-08	Compatibility	Latest two versions of Chrome, Edge, Firefox and Safari; responsive layout for phone browsers; JPEG/PNG	Tested on at least 3 browsers and 1 phone-size viewport
NFR-09	Maintainability	Modular code per barrier, documented API (OpenAPI), CI with automated tests	>= 70% unit-test coverage on core modules
NFR-10	Explainability	Every verdict shall be traceable to evidence from at least one barrier	100% of reports contain per-barrier evidence
NFR-11	Accessibility and localisation	Support Arabic (RTL) and English; readable font sizes and contrast	Checked against WCAG 2.1 AA contrast
NFR-12	Auditability	All verifications and admin actions logged with timestamp and actor	Log entries immutable by normal users

3.6 Security Threats Considered

Threat	Example	Planned control
Image modified after issuing	Change the name on a certificate	Watermark damage check (Barrier 1) plus tampering detection and ELA map (Barrier 2)
Watermark removal or screenshot laundering	Re-photograph the screen	AI detector; status "watermark absent" is not proof of authenticity
Watermark copied to another image	Paste a valid mark onto a forged image	Match payload with a database record; Barrier 2 still analyses the image; documented limitation
Fake / AI-generated face	Face swap on an ID photo	Barrier 2 detection, face-region analysis
Stolen API keys / accounts	Abuse of the verification API	Rate limiting, key rotation, RBAC
Interception of uploads	Man-in-the-middle on upload	TLS 1.2+, encrypted storage, retention limits
Adversarial perturbation	Noise to fool the classifier	Basic robustness tests; documented limitation

3.7 Constraints and Assumptions

- Academic timeline of 4 weeks and open-source tooling only.
- Models are trained on public datasets; performance on real-world forgeries may be lower and will be reported honestly.
- Initial support for still images only; video is future work.
- Results are decision-support evidence, not legal proof.

3.8 Traceability: Objectives to Requirements

Objective	Functional requirements	Non-functional requirements
O1 Watermark	FR-10 to FR-14	NFR-01, NFR-02
O2 AI detection	FR-20 to FR-24	NFR-02, NFR-10
O3 Fusion and report	FR-30 to FR-32	NFR-05, NFR-06, NFR-10
O4 Web client and API	FR-01, FR-02, FR-40 to FR-45	NFR-03, NFR-04, NFR-05, NFR-07, NFR-08, NFR-09, NFR-11, NFR-12

3.9 Acceptance Criteria (Examples)

- **AC-1:** Given an image issued by the system, when it is verified unchanged, then the verdict is Verified and the report shows the matching issuer record.
- **AC-2:** Given a watermarked image recompressed at JPEG Q70, when verified, then the watermark is reported intact or partially damaged and the payload bit-accuracy is at least 95%.
- **AC-3:** Given a face-swapped test image from the held-out set, when verified, then Barrier 2 returns a manipulation score above the decision threshold.
- **AC-4:** Given an unregistered genuine photo, when verified, then the verdict is Unverified, never Verified.
- **AC-5:** Given that one barrier is unavailable, when an image is verified, then the other barrier still runs and the report states which barrier failed.